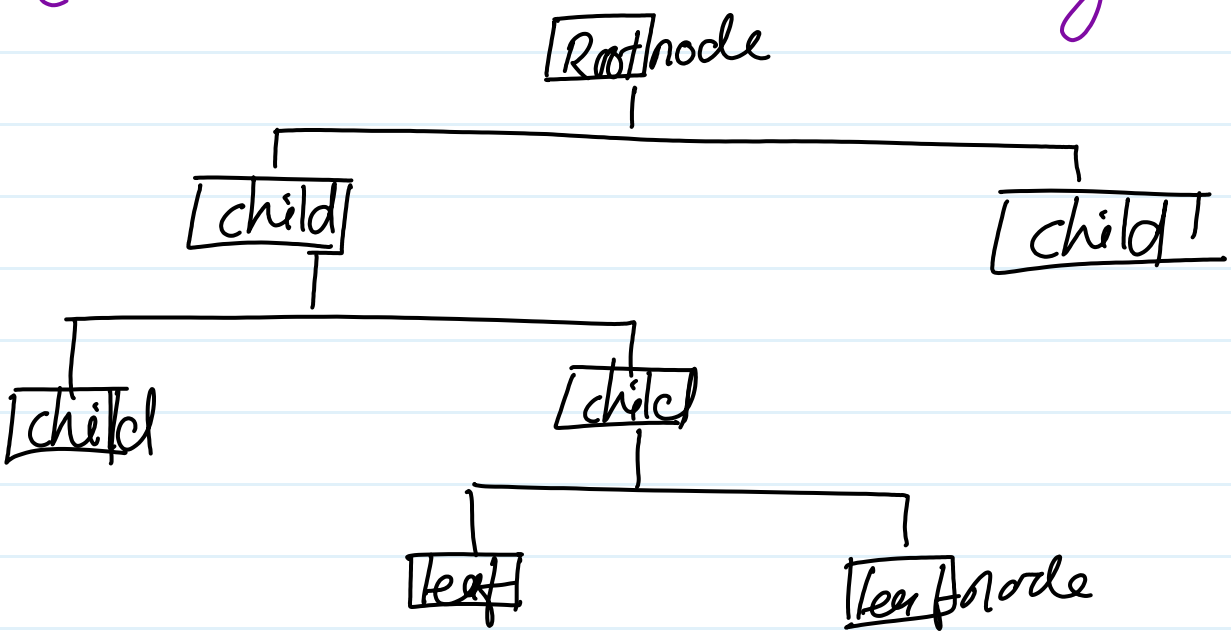
Decision tree

- classification
- Regression



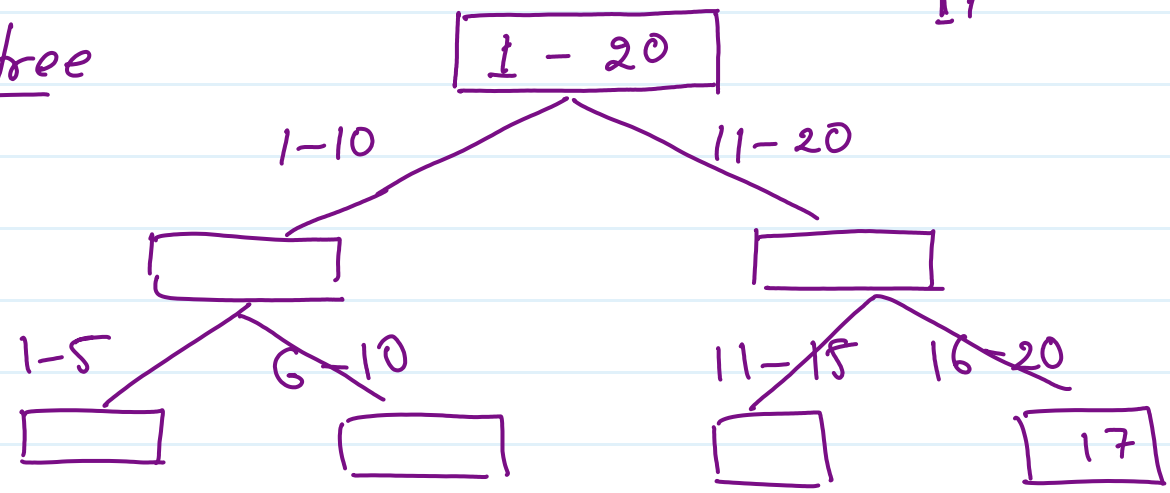
method of solve decision Tree algorithm –

1. ID3 — Entropy, Information Gain
2. Cart — Gini Index

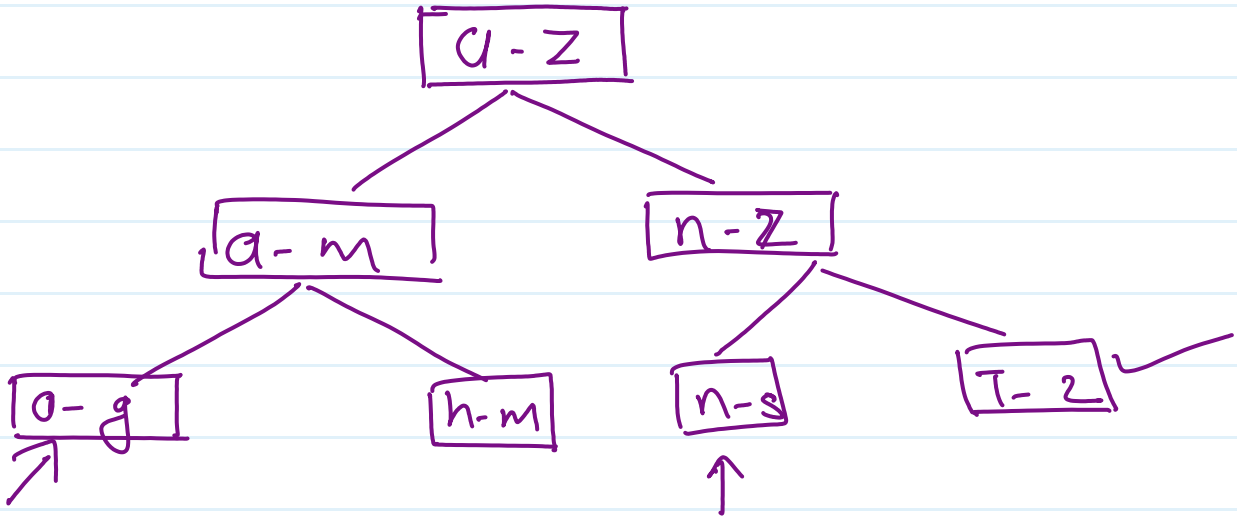
Dataset [5, 6, 3, 11, 10, 2, 3, 7, 8]

17

Binary tree



[Red, yellow, purple, blue, white, green, black]



Entropy OR Gini Index

Information Gain

* In DT we do not required to label categorical data into numerical

$x_1 \quad x_2 \quad x_3$

* Entropy and Gini Index → purity split in dataset

* Information Gain → DT feature split

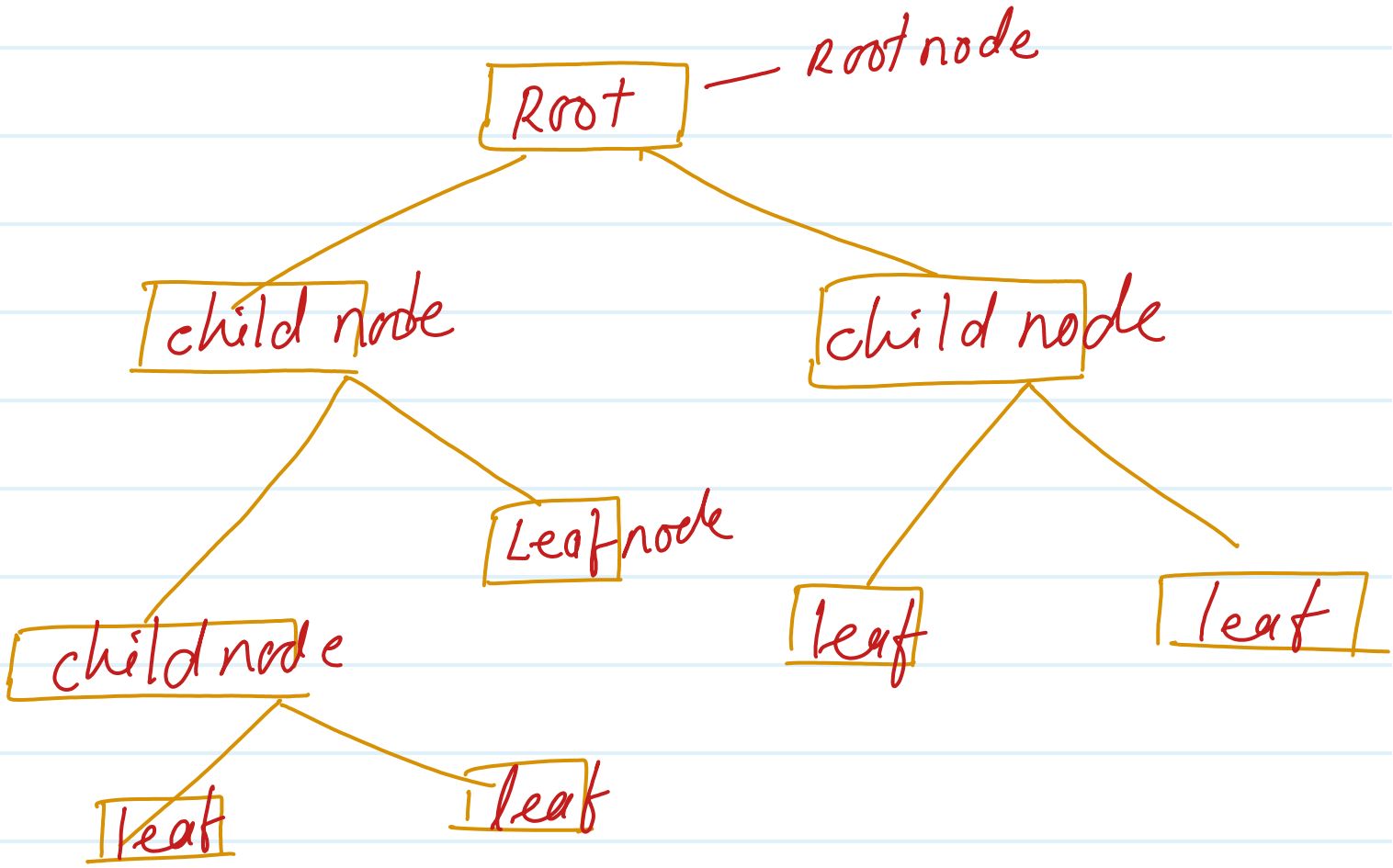
class.

weight	height	o/p obese/Noobese.
60	160	ob
70	170	NO
80	180	ob
90	190	NO
100	200	NO

(11) DT Regresser

Regression we use standard deviation / MSE / MAE

weight	height	BMI
60	160	21
70	170	22
80	180	20
85	190	23
90	195	24

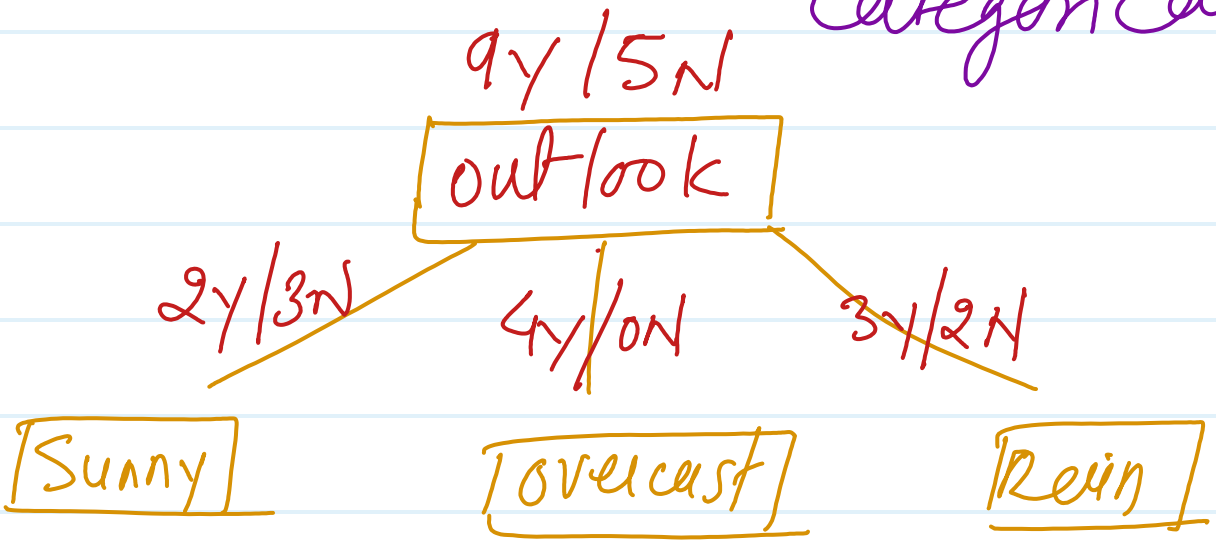


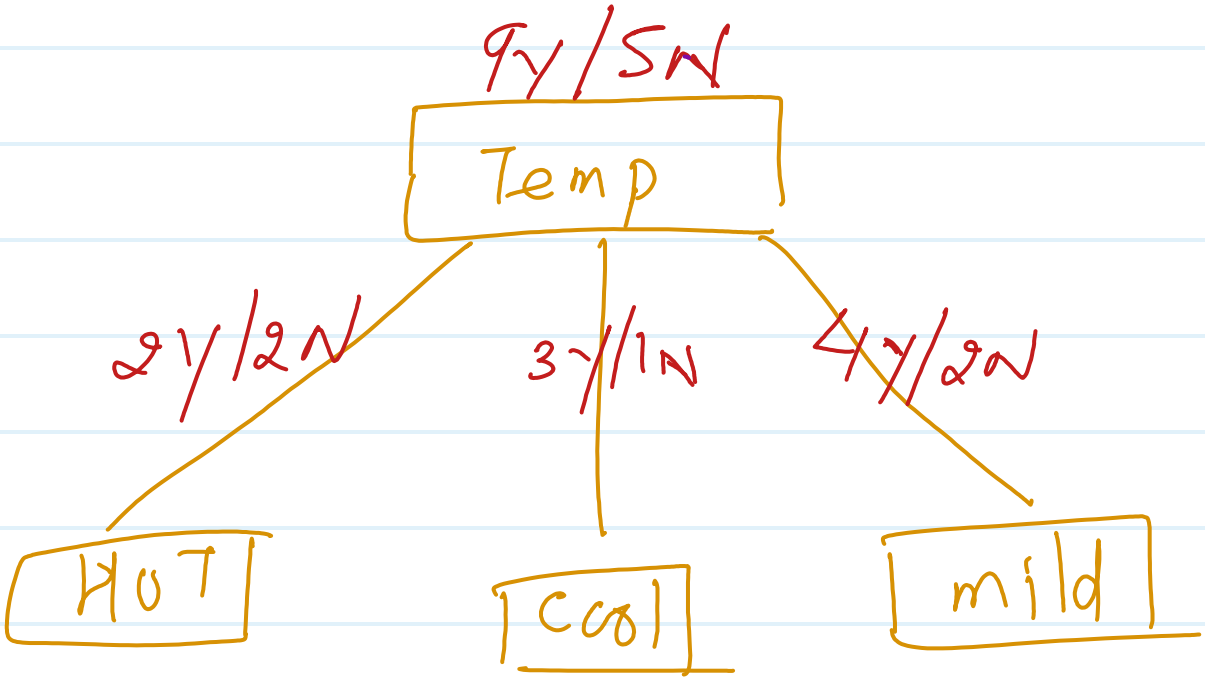
Decision Tree Classifier

outlook	Temp	humidity	wind	Tennis play
sunny	H	High	weak	N
sunny	H	H	strong	N
overcast	H	H	W	Y
rain	M	H	W	Y
rain	C	Normal	W	Y
rain	C	N	S	N
overcast	C	N	W	Y
sunny	M	H	W	N
sunny	C	N	W	Y
rain	M	N	W	Y
sunny	M	N	S	Y
overcast	M	high	S	Y
overcast	H	N	W	Y
rain	M	H	S	N

// Feature can be numeric and categorical

// Output can be numeric and categorical





ⓐ Entropy $H(s)$

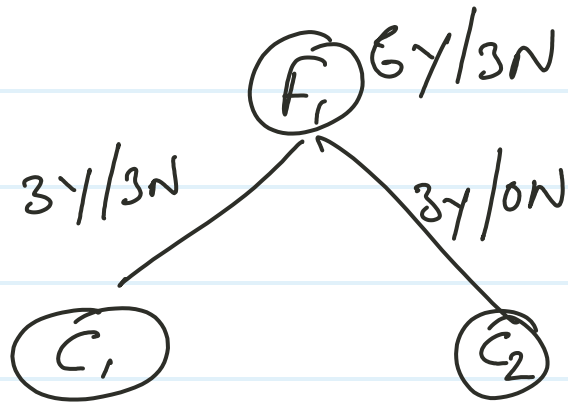
⇒ **Formula** (Binary class)

$$H(s) = -P_{yes}\log_2(P_{yes}) - P_{No}\log_2(P_{No})$$

multiclass

$$H(s) = -P_{C_1}\log_2(P_{C_1}) - P_{C_2}\log_2(P_{C_2}) - P_{C_3}\log_2(P_{C_3})$$

Example



F_1	F_2	F_3	O/P

$$C_1 \Rightarrow H(s) = -\frac{3}{6}\log\left(\frac{3}{6}\right) - \frac{3}{6}\log\frac{3}{6}$$

$\Rightarrow$ 1 impure split

$$C_2 \Rightarrow H(s) = -\frac{3}{3}\log\frac{3}{3} - \frac{0}{3}\log\frac{0}{3}$$

$\Rightarrow$ 0 pure split

For the pure split of feature

pure entropy should be zero (0)

for impure split = 1

(b) Gini index (Impurities).

main formula -

$$G.I. = 1 - \sum_{i=1}^{n} (P)^2$$

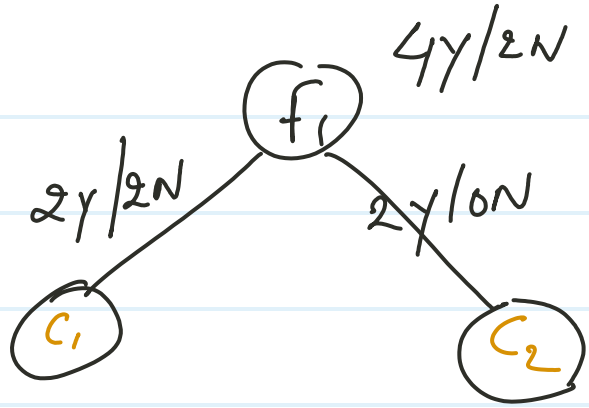
binary class.

$$G.I. = 1 - \sum_{i=1}^{n} \left[(P_{c_1})^2 + (P_{c_2})^2 \right]$$

multiclass

$$G.I. = 1 - \sum_{i=1}^{n} \left[(P_{c_1})^2 + (P_{c_2})^2 + (P_{c_3})^2 + \cdots \right]$$

Example



$C_1 \Rightarrow G.I. = 1 - \left[\left(\frac{2}{4}\right)^2 + \left(\frac{2}{4}\right)^2\right]$

$\boxed{= 0.5}$ ✓

$C_2 \Rightarrow G.I. = 1 - \left[\left(\frac{2}{2}\right)^2 + \left(\frac{0}{2}\right)^2\right]$

$\boxed{= 0}$ ✓

Range of entropy = 0 to 1

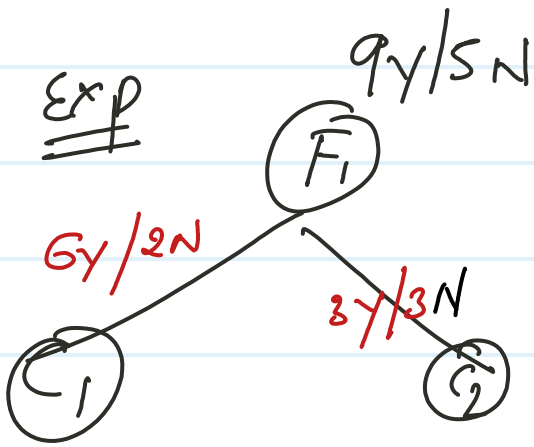
Gini impurity (index) = 0 to 0.5

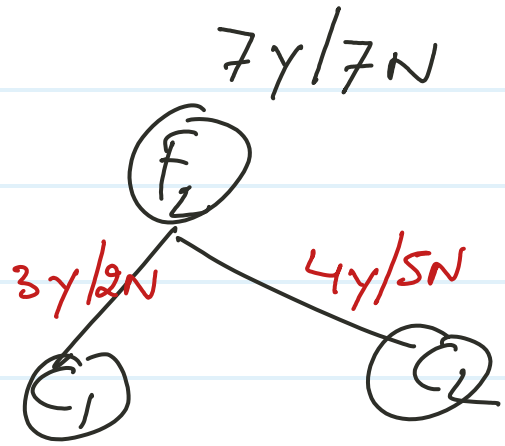
ⓒ Information Gain

Formula –

$$\text{gain}(S, f_1) = H(S) - \sum_{i=1}^{n} \frac{|S_v|}{|S|} H(S_v)$$

Exp





⇒ For F_1 ⇒

entropy $H(S) = -\frac{9}{14}\log\frac{9}{14} - \frac{5}{14}\log\frac{5}{14}$

$$\boxed{H(S) = 0.94}$$

$C_1 \Rightarrow H(S) = -\frac{6}{8}\log\frac{6}{8} - \frac{2}{8}\log\frac{2}{8}$

$$\boxed{H(S) = 0.81}$$

$C_2 \Rightarrow$

$$H(S) = -\frac{3}{3}\log\frac{3}{3} - \frac{0}{3}\log\frac{0}{3}$$

$$\boxed{H(S) = 1}$$

gain of $F_1 \Rightarrow$

$$\text{gain}(S, F_1) = 0.94 - \left[\frac{8}{14} \times 0.81 + \frac{6}{14} \times 1\right]$$

$$\text{gain}(S, F_1) = 0.049$$

$F_2 \rightarrow$ $$H(S) = -\frac{7}{7}\log\frac{7}{7} - \frac{7}{7}\log\frac{7}{7}$$

$$= 0$$

$C_1 \rightarrow$ $$H(S) = -\frac{3}{5}\log\frac{3}{5} - \frac{2}{5}\log\frac{2}{5}$$

$$= 0.133 + 0.159$$

$$\boxed{= 0.29}$$

$$C_2 \Rightarrow H(S) = -\frac{4}{9}\log\frac{4}{9} - \frac{5}{9}\log\frac{5}{9}$$

$$\boxed{= 0.014}$$

$$f_2 \quad \text{gain}(S, f_2) = 0 - \left[\frac{5}{14}\times 0.29 + \frac{9}{14}\times 0.014\right]$$

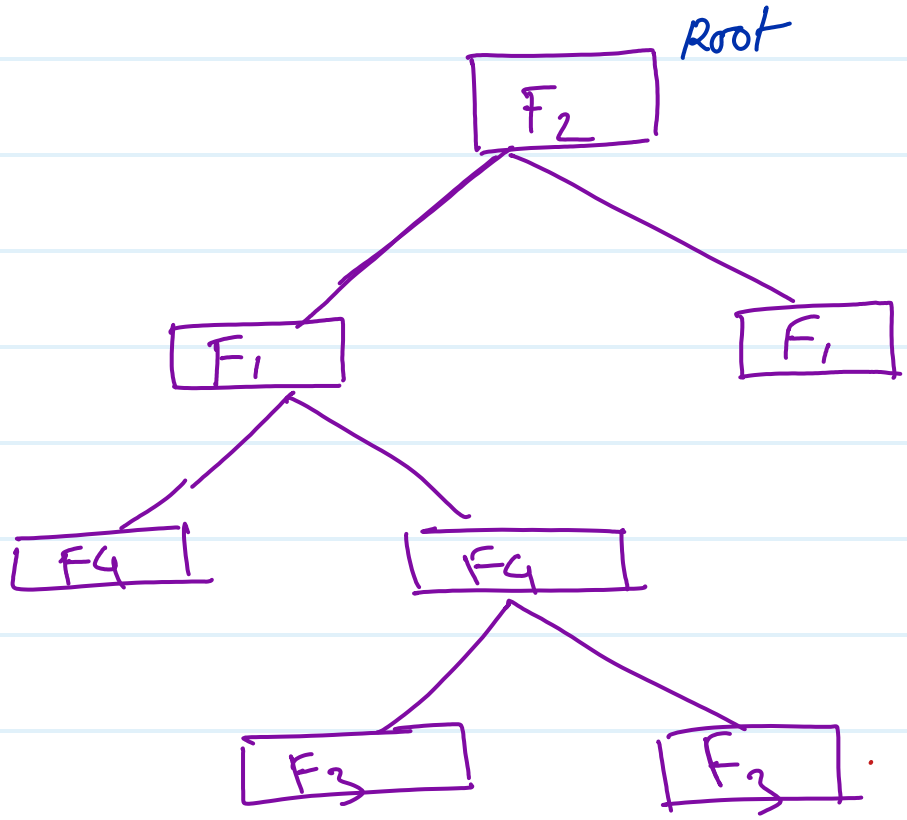
$$= 0 - [0.10 + 0.009]$$

$$\boxed{= -0.10}$$

F_1	F_2	F_3	F_4
0.49	0.56	0.025	0.10

Since F_2 has higher value of information gain's among the all feature so that it will be our root node.

F_1	F_2	F_3	F_4
0.49	0.56	0.025	0.10



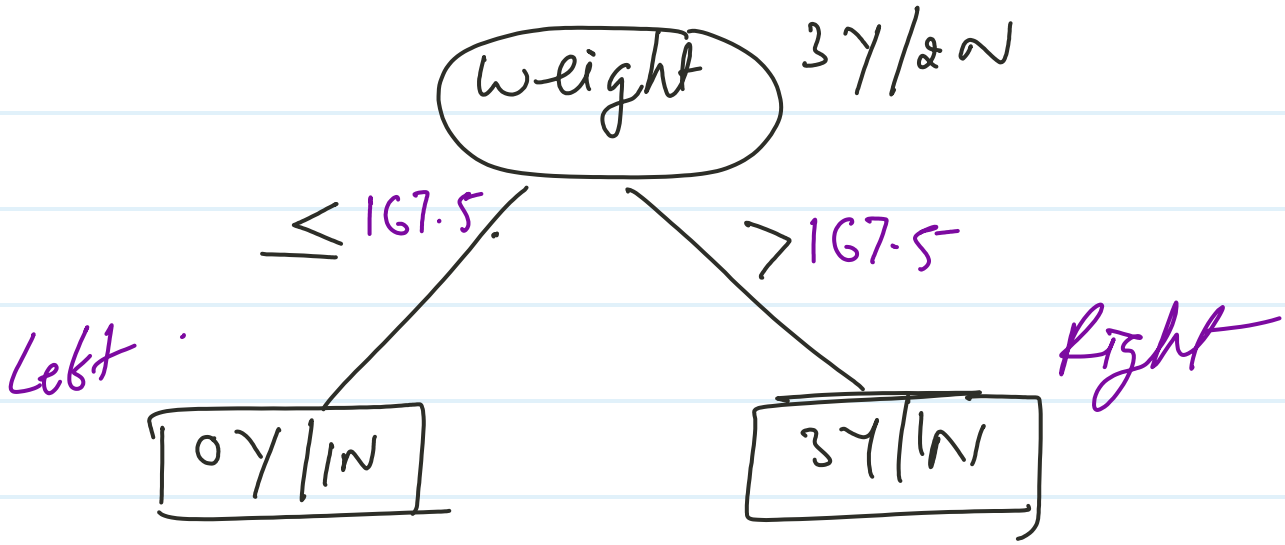
★ Independent analysis befor making DT

build DT with numerical feature

weight	heart De.
220	Y
180	Y
225	Y
190	N
155	N

weight		Heart
155		N
	167.5	
180		Y
	185	
190		N
	204	
220		Y
	222.4	
225		Y

with respect to every point avg. value need to find out gini index / Entropy



$$\text{Gini impurity} = 1 - \sum_{i=1}^{n} p_i^2$$

$$\text{gini (Left)} = 0$$

$$\text{gini (Right)} = 1 - \left[\left(\frac{3}{4}\right)^2 + \left(\frac{1}{4}\right)^2\right]$$

$$= 0.375$$

$$\text{Information gain} = G.I.[\text{Root}] - \sum_{\text{value}} \frac{|S_v|}{|S|} G.I.[\text{child}]$$

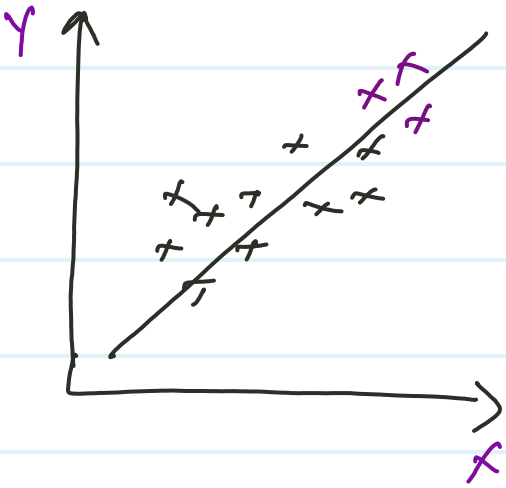
$$G.I.[\text{Root}] = 0.48$$

$$I.G.[167.5] = 0.48\left[\frac{1}{5} \times 0 + \frac{4}{5} \times 0.375\right]$$

$$I.G.[167.5] = 0.18$$

Information Gain should be high and Gini Index should be low.

✡ DT Regression



height		weight
160		50
	162.5	
165		65
	167.5	
170		85
	172.5	
175		70
	177.5	
180		90

height

≤ 162.5 $\quad$ > 162.5

$\boxed{50}$ $\quad$ $\boxed{65, 85, 70, 90}$

Left $\quad$ Right

50

mean = 77.5

heigh

≤ 167 $\quad$ > 167 $\quad$ splite

(65, 50) $\quad$ (90, 85, 70)

Ex

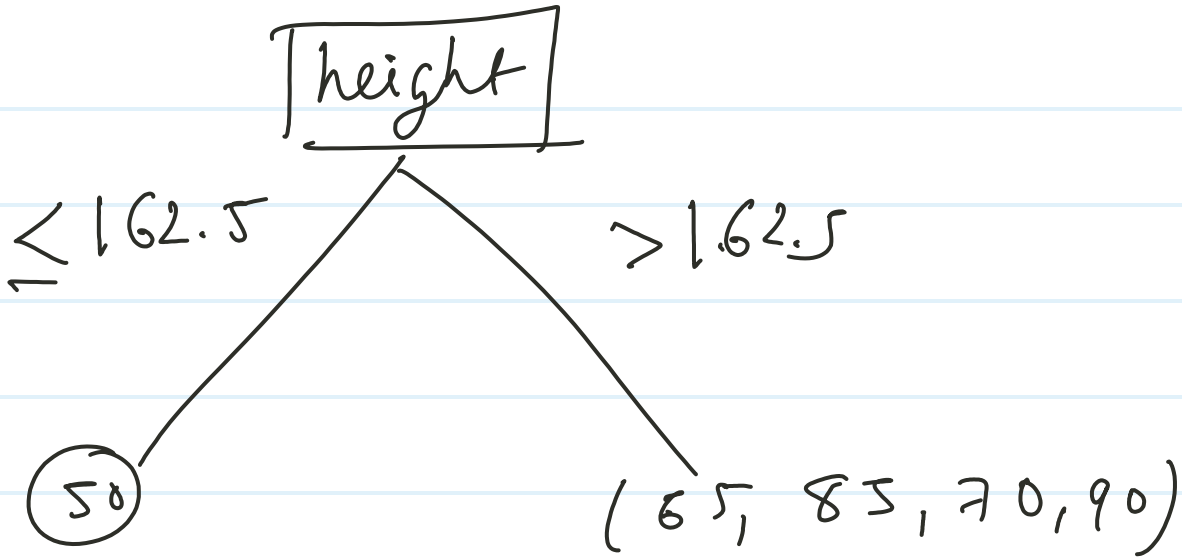
	height	weight	
	165	65	160
162.5			
	160	50	165
167.5			
	180	90	170
172.5			
	170	85	175
177.5			
	175	70	180

Regression problem weight calculated with respect to height

① Step - Short the value of height column (x Feature)

② step - Find Adjcent Avg. value b/w data point

③ step - Find Information gain with help of entropy and Gini Index.



mean = 77.5

Regression -

(1) mean = 77.5

(2) MSE, RMSE, MAE

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y - \hat{y})^2$$

$$\text{overall mean} = \frac{50 + 65 + 85 + 70 + 90}{5} = 72.$$

$$\text{height (variance)} = \frac{(72-50)^2 + (72-65)^2 + (72-85)^2 + (72-70)^2 + (72-90)^2}{5}$$

$\text{hight}_{(\text{variance})} = 206$

$$\text{var}(\text{Right}) = \frac{(77.5-65)^2 + (77.5-85)^2 + (77.5-70)^2 + (77.5-90)^2}{4}$$

$\text{var}(\text{Right}) = 106.25$

$\text{var}(\text{left}) = 50$

Reduction in variance

$$= \text{var}(\text{root}) - \sum_{i=1}^{n} w_i \times \text{var}[\text{child}]$$

$$= 206 - \left[\frac{1}{5} \times 0 + \frac{4}{5} \times 106.25\right]$$

Reduction variance $= 121$

we calculate MSE for all the datapoint whichever is less will be thresold..

height (x_1)	gender (x_2)	weight (y)
160	M	65
165	F	70
170	M	80
175	M	90
180	F	100

From height / Gender, choose root node

for height MSE = 55.5
for gender MSE = 53

so value of gender MSE is less
It will be our root node.

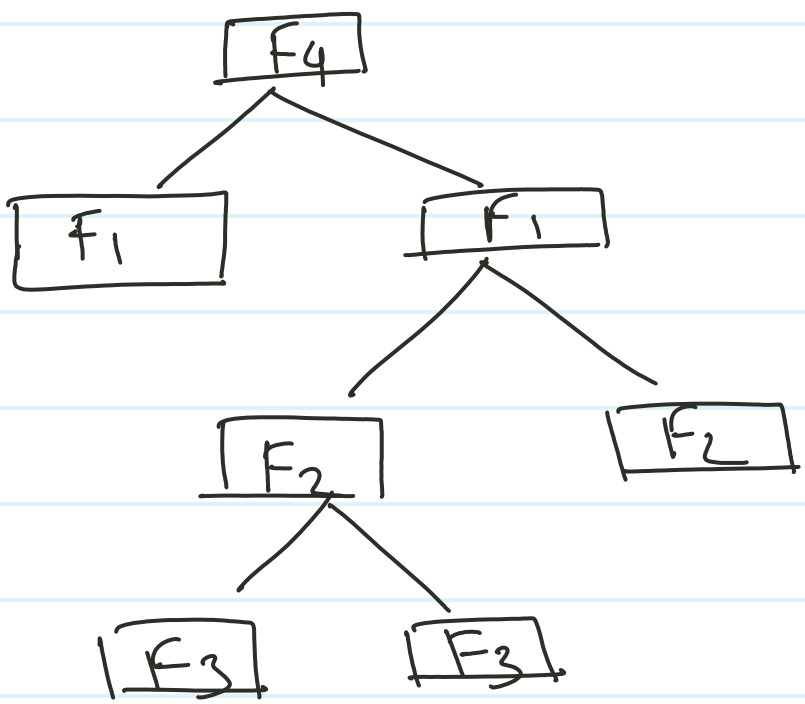
F_1	F_2	F_3	F_4	Target (y)
–	–	–	–	
–	–	–	–	
–	–	–	–	
–	–	–	–	

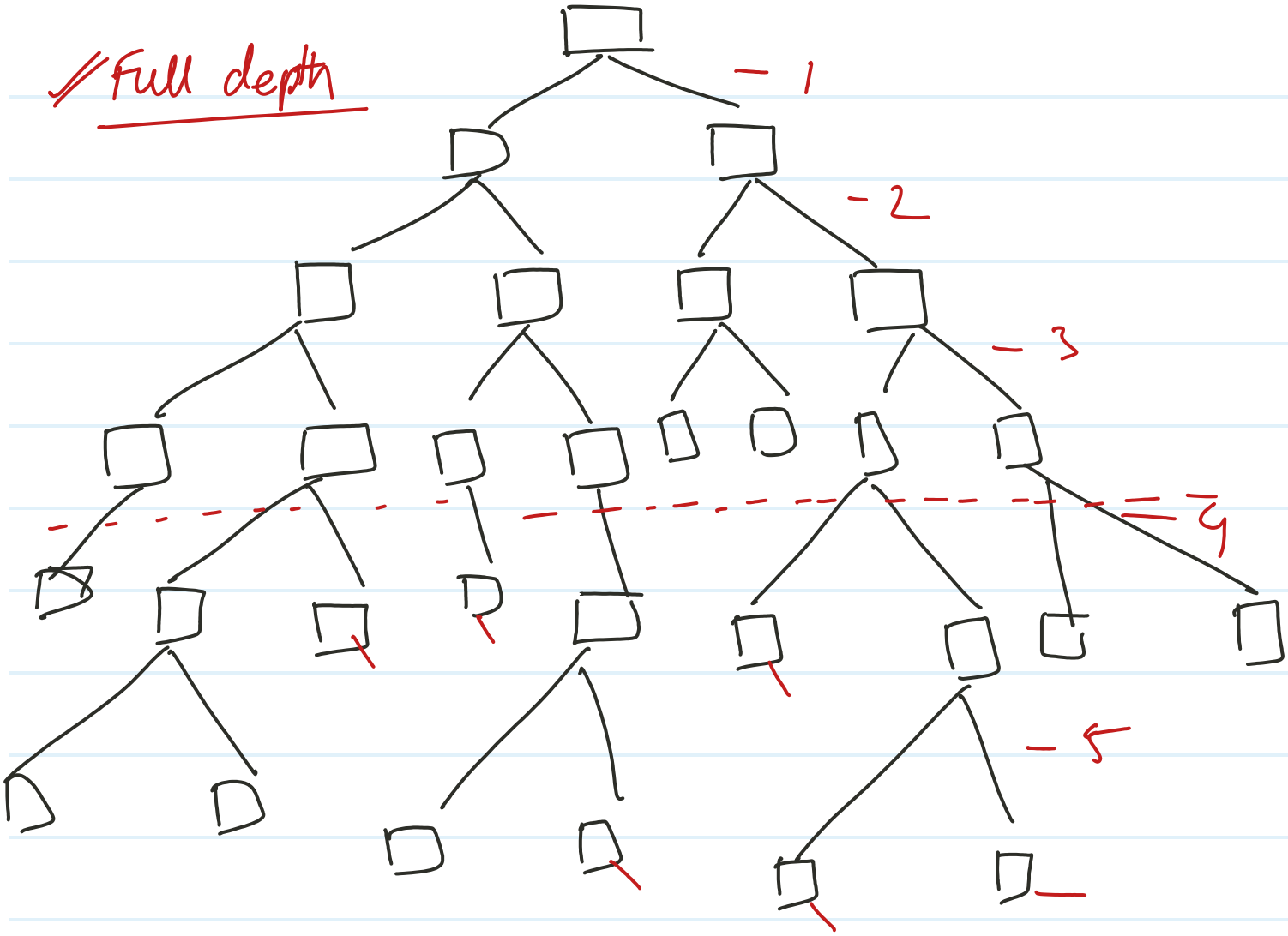
$F_1 \Rightarrow MSE = 49$

$F_2 \Rightarrow MSE = 55$

$F_3 \Rightarrow MSE = 60$

$F_4 \Rightarrow MSE = 43$





Pre-pruning and post-pruning

max-Depth = 5 [0, 5, 6, 8, 10]

mini-sample-leaf = 10 [5, 6, 8, 10, 12]

min-sample-split = 8 [4, 6, 8, 10, 12]

max-feature = 6 [2, 4, 6, 8, 10]

These 4 hyperparameters selected for pre-pruning before build DT. algorithms.

Post pruning =>

① make DT till end
② cut DT. using ccp value.
③ ccp value is nothing but threshold for gini / Entropy.

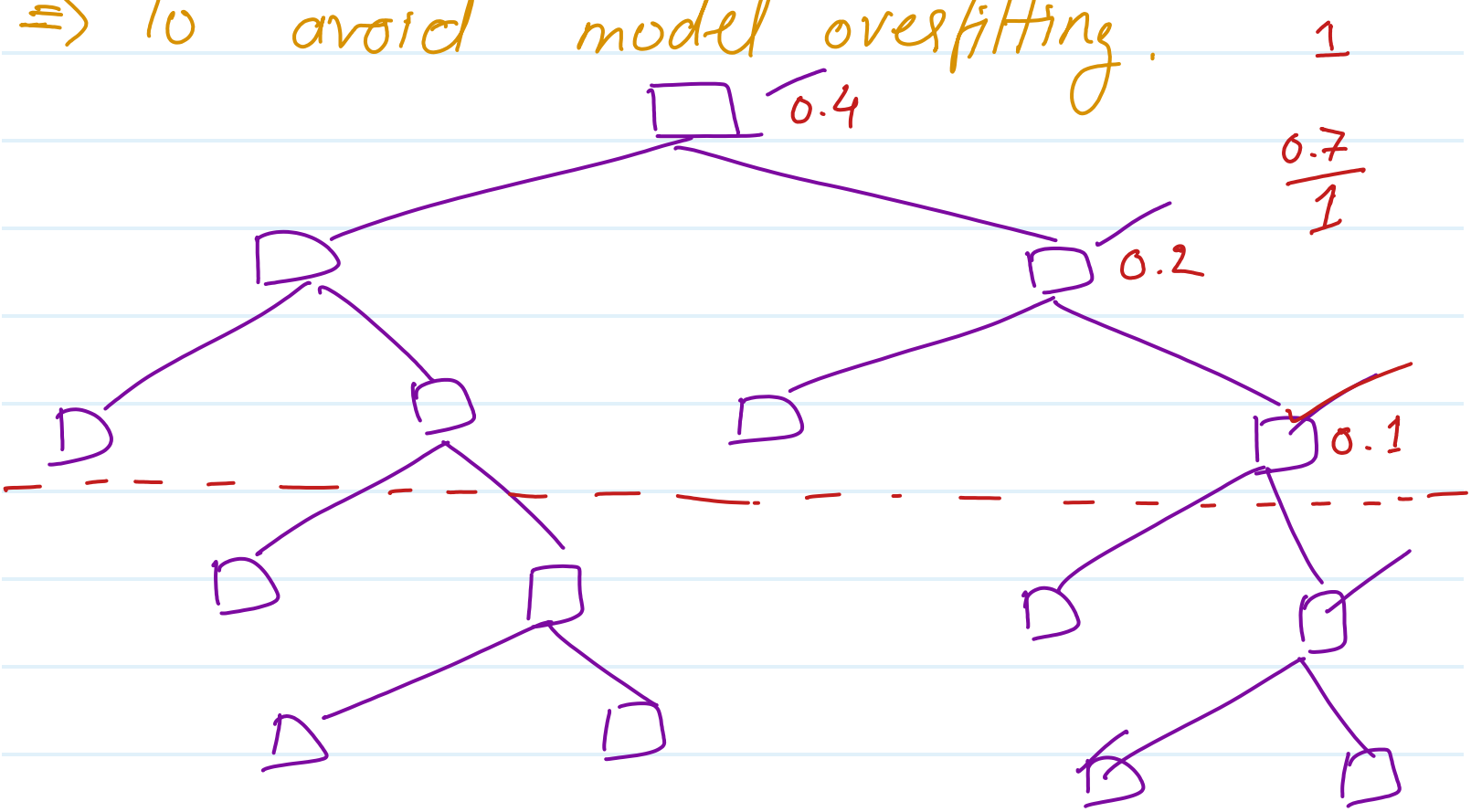
ccp value is responsible for depth of Tree.. If ccp is less, the depth will be less.
High ccp value the depth will be more.

ccp = [0.4, 0.5, 0.6, 0.01]

For model training either we can use pre-prunning or post-prunning.

① When we have large dataset at this time we use pre-prunning.

② When we have small dataset at this time we use postprunning.

Why we use post or pre-pruning?

⇒ To avoid model from overfitting.